ELECTIVE COURSE-1.2

UCS622: CONVERSATIONAL AI: ACCELERATED DATA SCIENCE [ADVANCED]				
	L	T	P	Cr
	2	0	2	3.0

Course Objective: This course will provide students with advanced methods of data science: Machine Learning with RAPIDS on Text and Graphical problems.

Syllabus

Introduction: Introduction to Machine Learning, Supervised and Unsupervised, Graph and Text Analytics, GPU Computing.

Introduction to Machine Learning - Supervised: Introduction to Supervised Learning, Linear Model, RAPIDS acceleration: Linear Regression, Overfitting and Cross Validation, Decision Tree, Visualizing Classification: {ROC, AUC, Confusion Matrix}, Bagging, Random Forests, RAPIDS Acceleration: Random Forest, Boosting, RAPIDS acceleration: K- NN, XGBoost.

Introduction to Machine Learning - Unsupervised: Introduction to Unsupervised Learning, Kmeans & Hierarchical Clustering, RAPIDS acceleration: K-Means, DBSCAN, PCA, t-SNE, UMAP, Visualizing Clusters, RAPIDS acceleration: PCA [t-SNE], UMAP, DBSCAN.

Graph Analytics: How to Represent & Store Graphs, Graph Power Laws, Centralities: Degree, Betweenness, Clustering Coefficient, PageRank & Personalized PageRank, Interactive Graph Exploration, RAPIDS Acceleration: Graphistry & cuXFilter.

Fundamentals of Text Analytics: Basics: Preprocessing, Representation, Word Importance, Latent Semantic Indexing, SVD: Dimensionality Reduction, Text Visualization.

Laboratory Work

- Decision Tree Classification Clustering in RAPIDS.
- Random Forest Classification in RAPIDS.

  {DLI Online Course Section: Fundamentals of Accelerated Data Science with RAPIDS, Section 2: GPU-accelerated Machine Learning}
- KMeans Clustering Implementation in RAPIDS.
- Dimensionality Reduction and Visualization in RAPIDS.
- Graph Analytics with cuGraph.
- Latent semantic indexing for text via singular value decomposition(cuML).
- Accelerating Workloads using RAPIDS

  {**DLI Online Course: Fundamentals of Accelerated Data Science with RAPIDS**}

Course Learning Outcomes (CLOs)

The students will be able to:

1. analyze methods and theories in the field of machine learning and provide an introduction to the basic principles, techniques, and applications of machine learning, supervised, unsupervised and reinforcement learning.
2. comprehend and apply different classification and clustering techniques.
3. understand GPU computing for building advanced data science applications.
4. deep understanding Machine Learning, Data Analytics and Data Science Toolkits with optimized acceleration using RAPIDS Framework.

Text Books

1. Mitchell M., T., Machine Learning, McGraw Hill (1997) 1st Edition.
2. Alpaydin E., Introduction to Machine Learning, MIT Press (2014) 3rd Edition.
3. Vijayvargia Abhishek, Machine Learning with Python, BPB Publication (2018).

Reference Books

1. Bishop M., C., Pattern Recognition and Machine Learning, Springer-Verlag (2011) 2nd Edition.
2. Michie D., Spiegelhalter J. D., Taylor C. C., Campbell, J., Machine Learning, Neural and Statistical Classification. Overseas Press (1994).

Evaluation Scheme:

SI. No	Evaluation Elements	Weightage (%)
1	MST	35
2	EST	35
3	Sessional (May include Assignments/Projects/Tutorials/Quiz/Lab evaluations)	30